ALICE L WALTON SCHOOL OF MEDICINE

ST Segment Changes: Recognition and Localization of Myocardial Ischemia

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Describe and define the ST segment
2. Recognize the localization and types of ST changes
3. Compare common ST segment patterns and their implications for ischemia localization

CASE CONNECTION

A 58-year-old male presents to the emergency department with acute-onset chest pain lasting 45 minutes. During a cardiology rotation, medical student Z.M. is asked to review the patient's ECG and explain the findings, focusing on the ST segment.

What changes might be seen in the ST segment, and what can they reveal about myocardial ischemia localization? Consider your answers as you read, and we'll revisit this case at the end of the brick.

GO TO CONCLUSION ↓

Definition and Boundaries (LO#1)

The ST segment is defined as the portion of the ECG tracing that extends from the end of the QRS complex (J point) to the beginning of the T wave (Figure 1).

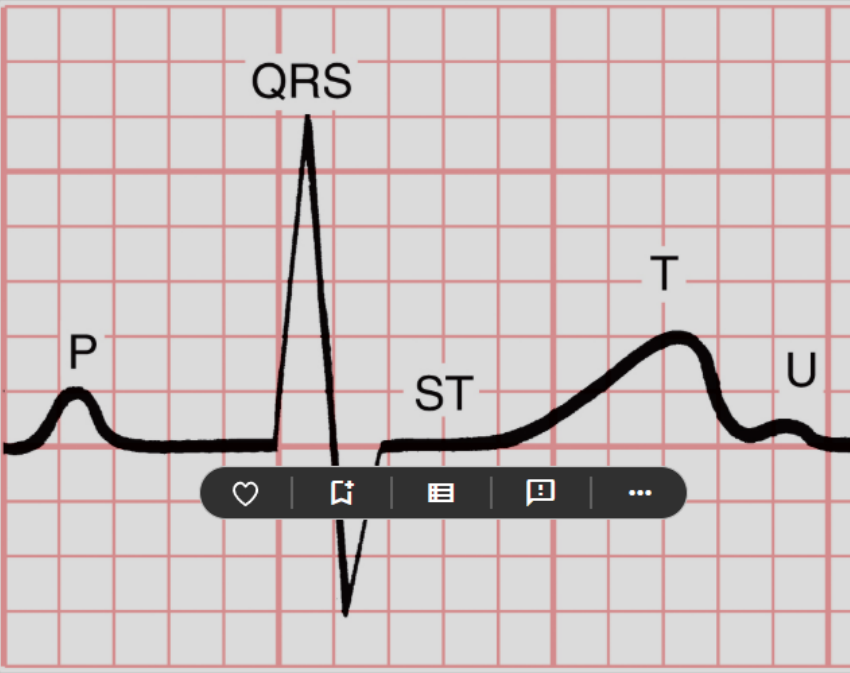


Figure 1. Waves and segments of the electrocardiogram

Note that the ST segment in the figure above is isoelectric. That is, it is even with the baseline of the recording. This is the normal state. The ST segment corresponds to the plateau phase (phase 2) of the ventricular action potential, when both ventricles are completely depolarized. This is shown in Figure 2 below.

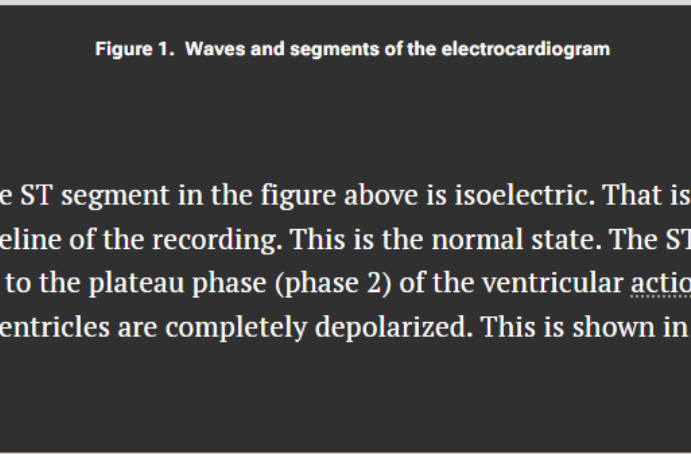

Action potential in contractile cell
ECG recorded on body surface
0
1
2
3
4
ST segment

Figure 2. The ST segment on e) corresponds to phase 2 of the action potential (recorded from inside a cell).

What defines the ST segment on an ECG and where is it located?

ST segment elevation and depression

In cases of ischemia, the ST segment can deviate below or above the baseline (ST segment depression or ST segment elevation, respectively) as shown in Figure 3 below.

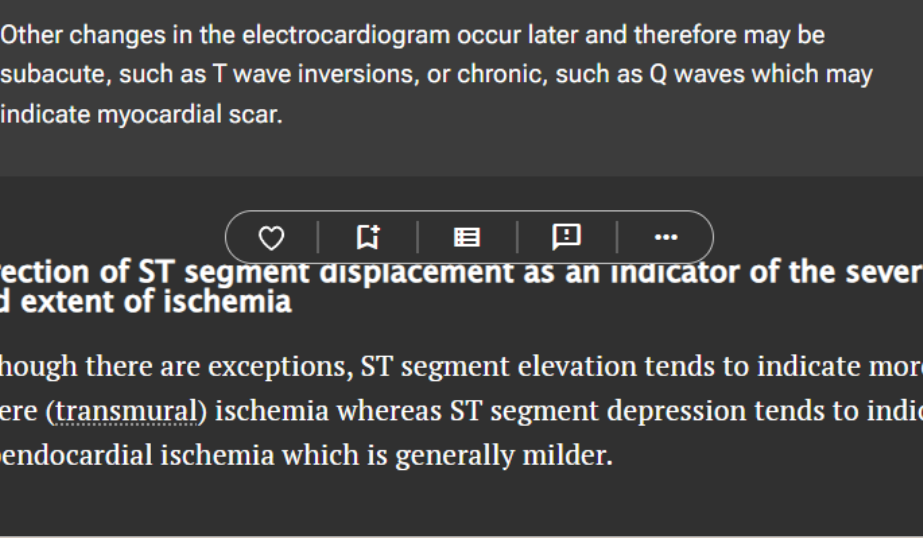


Figure 3. ST depression (top) and ST elevation (bottom)

Importance of the ST segment in assessing the acuity of myocardial infarction

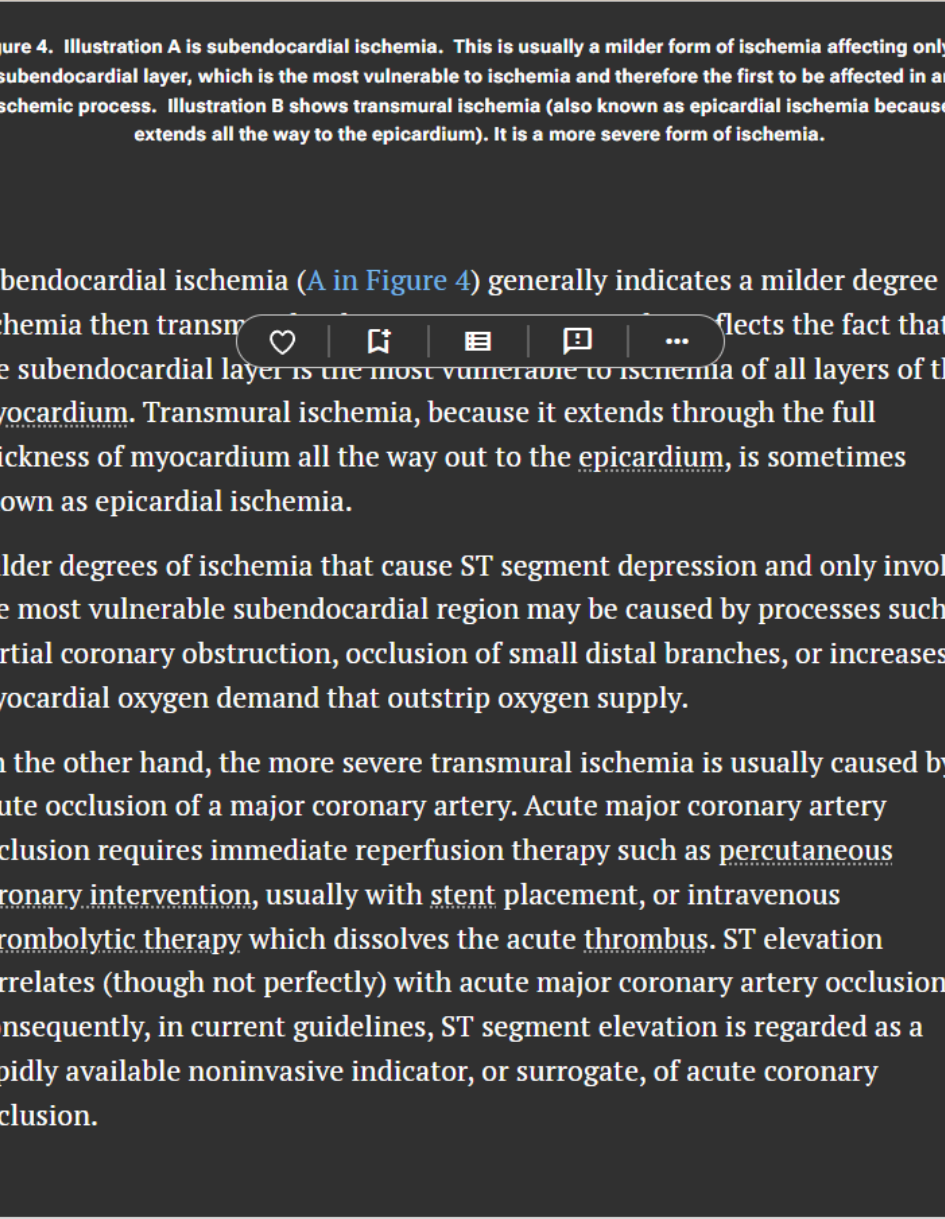
In acute myocardial ischemic syndromes, the ST segment is often the first portion of the electrocardiogram to change, generally within minutes. Thus, the ST segment indicates the *acuity* of the infarction and hence the *immediacy* of need for treatment.

CLINICAL CORRELATION

Other changes in the electrocardiogram occur later and therefore may be subacute, such as T wave inversions, or chronic, such as Q waves which may indicate myocardial scar.

Direction of ST segment displacement as an indicator of the severity and extent of ischemia

Although there are exceptions, ST segment elevation tends to indicate more severe (transmural) ischemia whereas ST segment depression tends to indicate subendocardial ischemia which is generally milder.


Subendocardial Injury: ST Depression
ST
V5
A
Transmural (Epicardial) Injury: ST Elevation
ST
V5
B

With subendocardial ischemia the ST segment is depressed.

With epicardial (transmural) ischemia the ST segment is elevated.

These are general, though not absolute, rules.

Image source: Braunwald's Heart Disease: A Textbook of Cardiovascular Medicine

Figure 4. Illustration A is subendocardial ischemia. This is usually a milder form of ischemia affecting only the subendocardial layer, which is the most vulnerable to ischemia and therefore the first to be affected in any ischemic process. Illustration B shows transmural ischemia (also known as epicardial ischemia because it extends all the way to the epicardium). It is a more severe form of ischemia.

Subendocardial ischemia (A in Figure 4) generally indicates a milder degree of ischemia then transm flects the fact that the subendocardial layer is the most vulnerable to ischemia of all layers of the myocardium. Transmural ischemia, because it extends through the full thickness of myocardium all the way out to the epicardium, is sometimes known as epicardial ischemia.

Milder degrees of ischemia that cause ST segment depression and only involve the most vulnerable subendocardial region may be caused by processes such as partial coronary obstruction, occlusion of small distal branches, or increases in myocardial oxygen demand that outstrip oxygen supply.

On the other hand, the more severe transmural ischemia is usually caused by acute occlusion of a major coronary artery. Acute major coronary artery occlusion requires immediate reperfusion therapy such as percutaneous coronary intervention, usually with stent placement, or intravenous thrombolytic therapy which dissolves the acute thrombus. ST elevation correlates (though not perfectly) with acute major coronary artery occlusion. Consequently, in current guidelines, ST segment elevation is regarded as a rapidly available noninvasive indicator, or surrogate, of acute coronary occlusion.

ACTIVE LEARNING QUESTION

A 65-year-old patient presents with chest pain. The ECG shows 3 mm ST elevation in leads V2-V4. What does this finding most likely indicate?

- A. ○ Unstable angina
- B. ● Acut
- C. ○ Acute pericarditis
- D. ○ Ventricular aneurysm
- E. ○ Early repolarization

Explanation

The correct answer is acute anterior STEMI (B). ST elevation ≥2 mm in contiguous precordial leads V2-V4 indicates acute transmural myocardial infarction in the anterior wall territory. Unstable angina (A) typically shows ST depression or T wave changes. Pericarditis (C) usually shows diffuse ST elevation with PR depression. Ventricular aneurysm (D) shows chronic, stable ST elevation. Early repolarization (E) shows concave ST elevation, typically <2 mm.

Localization of ischemia (LO#2)

The relationship between specific ECG leads and corresponding myocardial regions forms the basis for localizing ischemia and infarction using ST segment changes. This correlation between leads and anatomic myocardial regions is helpful for determining the culprit vessel and guiding therapies such as percutaneous coronary intervention.

First, we must apprec ly the portions of the myocardium (sometimes referred to as regions or walls) supplied by the various arteries and their branches.

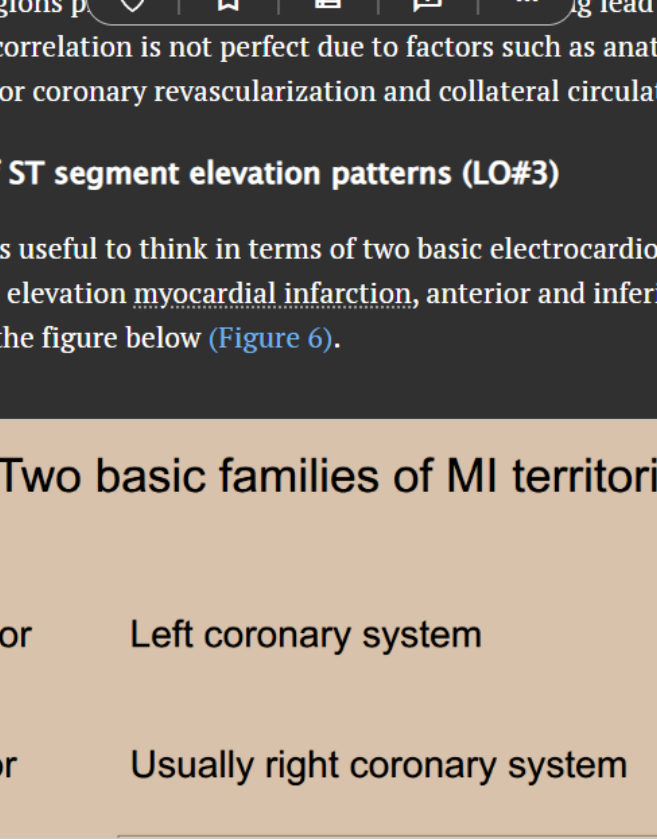

Left main coronary artery
Left circumflex coronary artery
Aorta
Right coronary artery
Left anterior descending coronary artery

Figure 5. Coronary anatomy

The left (left main) coronary artery divides into the left anterior descending and circumflex coronary arteries and their branches. As illustrated in Figure 5, they supply the anterior and lateral walls of the left ventricle, respectively. The right coronary artery supplies the inferior and posterior walls of the left ventricle as well as the right ventricle. Transmural ischemia in these myocardial regions p g lead groups although the correlation is not perfect due to factors such as anatomic variations, prior coronary revascularization and collateral circulation.

"Families" of ST segment elevation patterns (LO#3)

In general, it is useful to think in terms of two basic electrocardiographic patterns of ST elevation myocardial infarction, anterior and inferior, as illustrated in the figure below (Figure 6).

Two basic families of MI territories

Anterior	Left coronary system
Inferior	Usually right coronary system

Figure 6. STEMI pattern groups, or "families"

An important exception to the figure above is that occlusions of the circumflex coronary artery, considered to be in the left coronary system, can sometimes produce inferior infarctions.

Included in the inferior infarction "family" are right ventricular and posterior infarctions. Detectio dvanced electrocardiographic techniques which are discussed toward the end of this brick.

Correlation of lead groups with vascular territories

Different leads, or groups of leads, correspond to the various territories. Identification of which leads contain ST segment elevations can help localize the ischemic process to a particular myocardial region which may correspond to the culprit artery and thus help guide intervention.

Correspondence between ECG lead groups and myocardial (and vascular) regions is illustrated in the figures below. The lead groups listed are those which display ST elevations when the corresponding regions undergo transmural ischemia and infarction. Thus, ST segment *elevation* patterns are useful in localization of ischemia and infarctions. The anterior family is generally corresponding to the left coronary artery and its branches including left anterior descending (LAD) and circumflex arteries (Figure 7, Figure 8).

Anterior family

Anteroseptal	V1-V3
Anterolateral	Any of V1-V3 + V4-V6
Lateral wa	naybe V5-V6

Figure 7. Leads corresponding to types of anterior infarction

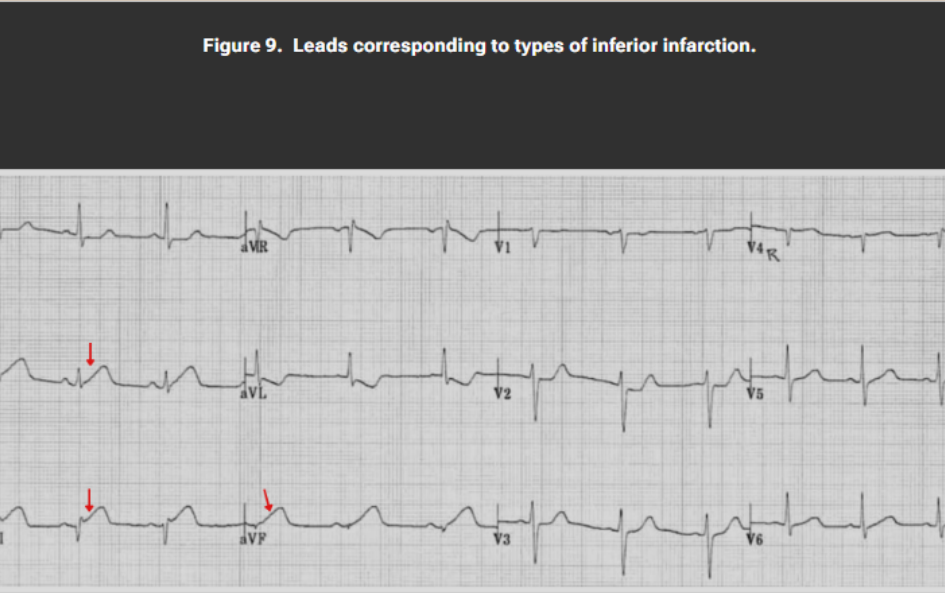


Figure 8 Anterolateral STEMI. Note STE in V2-5, I, aVL.

The inferior family is generally corresponding to the right coronary artery (RCA) and its branches (Figure 9, Figure 10).

Inferior family lead groups

Inferior infarct: II, III, AVF

Posterior infarct: Mirror image changes in V1-2 (eg tall R in V1 as a Q equivalent). M or infarct changes in II, III and AVF.

RV infarct: ST elevation may be present in V1 acutely. ST elevation in inferior leads (II, III, AVF) present acutely. A clue to RV involvement is when it is mainly in lead III. (Why??)

Figure 9. Leads corresponding to types of inferior infarction.

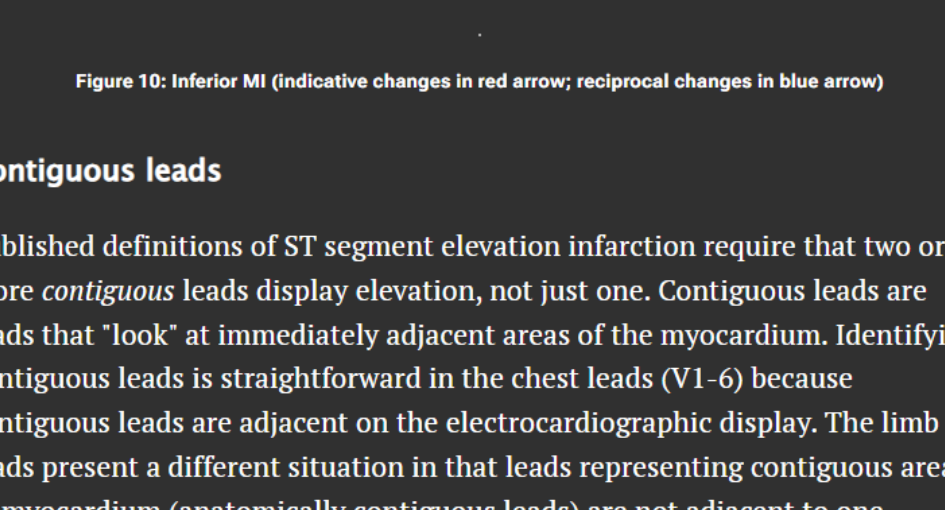


Figure 10: Inferior MI

Indicative and reciprocal ST segment changes

ST segment elevations tend to *indicate* the area of transmural infarction and therefore are referred to as *indicative* changes. Secondary ST segment changes (generally ST depressions) often occur in leads outside the infarct zone. These are known as *reciprocal* ST segment changes. Reciprocal changes can be seen in Figures 9 and 10 abo III, which is outside the infarct zone, shows reciprocal ST depression. In Figure 10 reciprocal STD is seen mainly in lead aVL.

Figure 10: Inferior MI (indicative changes in red arrow; reciprocal changes in blue arrow)

Contiguous leads

Published definitions of ST segment elevation infarction require that two or more *contiguous* leads display elevation, not just one. Contiguous leads are leads that "look" at immediately adjacent areas of the myocardium. Identifying contiguous leads is straightforward in the chest leads (V1-6) because contiguous leads are adjacent on the electrocardiographic display. The limb leads present a different situation in that leads representing contiguous areas of myocardium (anatomically contiguous leads) are not adjacent to one another on the electrocardiographic display. For example, anatomically contiguous leads representing the inferior wall, II, III and aVF, are not adjacent to one another on the electrocardiographic display. This is due entirely to tradition and convention, reflecting the historic order in which the leads were developed.

Treatment Implications

The clinical implications of ST segment patterns extend beyond diagnosis to treatment decisions and prognosis. ST elevation myocardial infarction (STEMI) requires immediate reperfusion therapy, either primary percutaneous coronary intervention or fibrinolytic therapy, with the goal of achieving reperfusion within 90-120 minutes of first medical contact.

Non-ST elevation patterns, including ST depression and T wave changes, indicate non-ST elevation acute coronary syndrome (NSTE-ACS), which encompasses unstable angina and non-ST elevation myocardial infarction. These patients require risk stratification to determine the timing and intensity of intervention. The presence and severity of ST depression help guide decisions about early invasive versus conservative management strategies.

ACTIVE LEARNING QUESTION

ST elevation in leads II, III, and aVF with reciprocal depression in I and aVL most likely indicates which type of myocardial infarction?

- A. ○ Anterior STEMI
- B. ○ Lateral STEMI
- C. ● Inferior STEMI
- D. ○ Posterior STEMI
- E. ○ Septal STEMI

Explanation

The correct answer is inferior STEMI (C). ST elevation in leads II, III, and aVF indicates inferior wall myocardial infarction, typically due to RCA occlusion d aVL supports this diagnosis. Anterior STEMI (A) would show changes in V1-V6. Lateral STEMI (B) affects I, aVL, V5-V6. Posterior STEMI (D) shows ST depression in V1-V3 with tall R waves. Septal STEMI (E) primarily affects V1-V2.

Advanced localization techniques

Advanced localization techniques include examining posterior leads (V7-V9), for posterior infarction and checking right-sided leads (V3R-V6R) for right ventricular involvement, both of which can accompany inferior infarction. These leads, depicted in the figures below, are not routinely placed for recording ECGs.

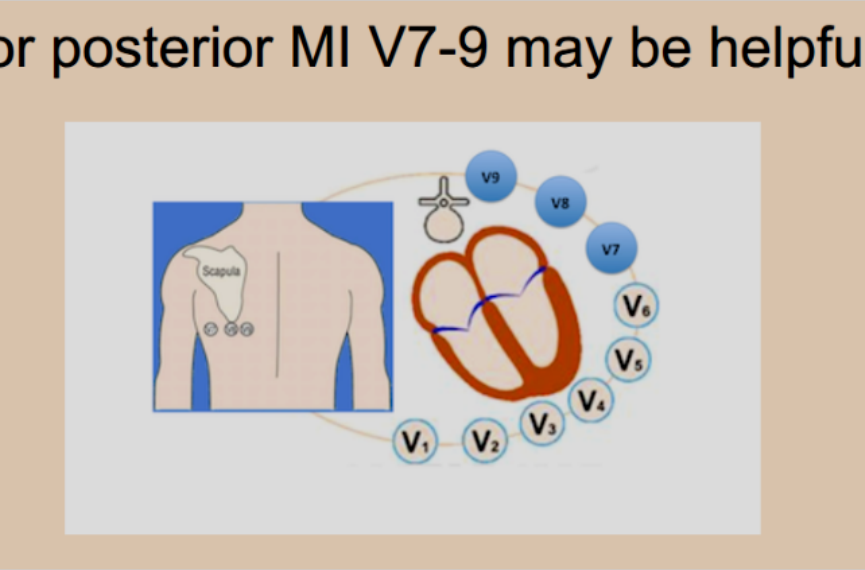

V3R and V4R

Image source: http://www.iepmoinhos.com.br/protocolos/publico/protocolos/informacoesQuadrinho/5912

Figure 12. Depiction of the chest lead placement, with V3r and V4r added for detection of acute right ventricular infarction.

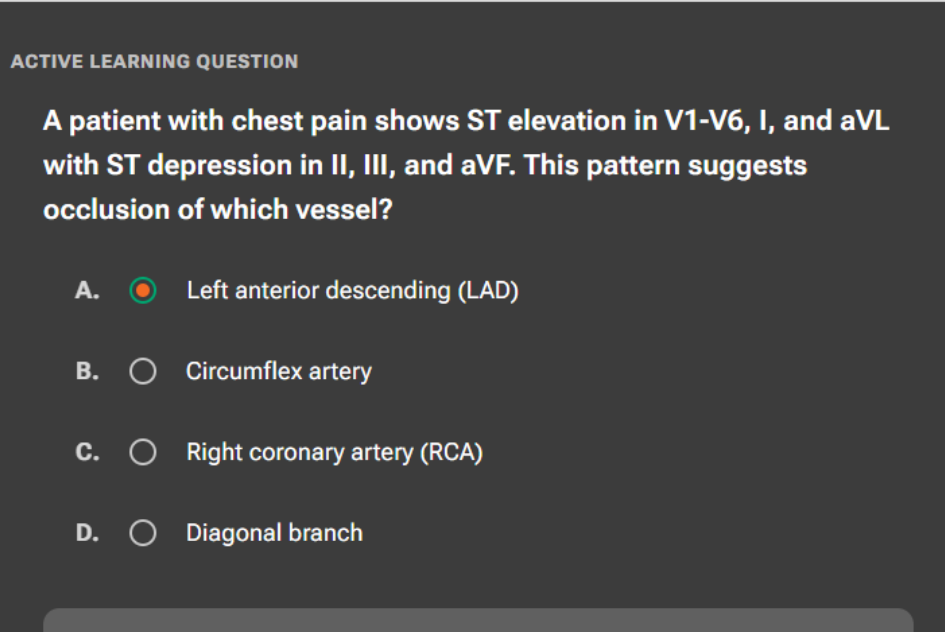

For posterior MI V7-9 may be helpful

Figure 13. Leads V7-9 added to the posterior thorax for detection of acute posterior infarction.

CLINICAL CORRELATION

ST elevation is a sur imperfect one. Around 30% of acute eperfusion will exhibit only subtle ECG findings which do not meet published ST elevation criteria. Identification of such patterns requires an advanced level of ECG interpretation which is beyond the scope of this brick.

ACTIVE LEARNING QUESTION

A patient with chest pain shows ST elevation in V1-V6, I, and aVL with ST depression in II, III, and aVF. This pattern suggests occlusion of which vessel?

- A. ● Left anterior descending (LAD)
- B. ○ Circumflex artery
- C. ○ Right coronary artery (RCA)
- D. ○ Diagonal branch

Explanation

The correct answer is Left anterior descending (LAD) (A). Extensive ST elevation across V1-V6 with involvement of I and aVL indicates a large anterior STEMI due to proximal LAD occlusion before the first septal and diagonal branches. The reciprocal ST depression in inferior leads supports this diagnosis. Left circumflex (B) affects lateral leads. Right coronary arter nal branch (D) would show more limited lateral involvement.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to the 58-year-old male, what ST segment changes did you expect on his ECG if he had anterior versus inferior ischemia? How do these changes help localize the site of myocardial injury?

If Z.M. observed ST elevation in leads V2-V4, this would indicate anterior wall ischemia due to LAD occlusion. Alternatively, ST elevation in leads II, III, and aVF with reciprocal depression in I and aVL would suggest inferior wall ischemia from RCA occlusion. The specific lead involvement allows precise localization of the culprit vessel and guides immediate reperfusion strategies.

Summary

- The ST segment on an ECG extends from the J point to the beginning of the T wave and is normally isoelectric, representing the plateau phase of ventricular depolarization.
- ST segment elevation typically indicates acute transmural myocardial injury, while ST depression suggests subendocardial ischemia.
- Localization of ischemia is achieved by correlating ST segment changes in specific ECG leads with myocardial regions, aiding in identifying the culprit coronary artery.
- Anterior wall infa sts LAD involvement, while inferior wall changes in leads II, III, aVF indicate RCA occlusion.
- Reciprocal ST depression occurs in leads outside the primary injury site and supports the diagnosis of acute transmural infarction.
- Systematic analysis of ST changes across contiguous leads is crucial for localizing ischemia and guiding targeted intervention for myocardial infarction.

Review Questions

1. What is the ST segment on an ECG and its place in the cardiac cycle?

- A. ○ From P wave to QRS start
- B. ○ Interval before T wave peak
- C. ● From J point to T wave start
- D. ○ Duration of PR interval
- E. ○ Between QRS start and next P wave

Hide Explanation

The correct answer is C. The ST segment (C) extends from the J point to the start of the T wave and represents the plateau phase of ventricular depolarization, normally appearing isoelectric tion, (B) describes the T wave, and (D) represents the PR interval, while (E) encompasses multiple parts of the cardiac cycle.

2. ST elevation in leads II, III, and aVF with reciprocal depression in I and aVL most likely localizes to which myocardial region?

- A. ○ Anterior wall
- B. ○ Lateral wall
- C. ● Inferior wall
- D. ○ Posterior wall
- E. ○ Septal wall

Hide Explanation

The correct answer is Inferior wall infarction (C) which corresponds to ST elevation in II, III, and aVF with reciprocal depression in I and aVL, typically due to RCA occlusion. (A) Anterior STEMI affects V1-V6. (B) Lateral STEMI involves leads I, aVL, V5, V6. (D) Posterior infarction shows ST depression in V1-V3. (E) Septal STEMI affects V1-V2.

3. What ST electrocardiographic pattern suggests acute coronary occlusion (transmural infarct needing immediate reperfusion)?

- A. ● ST segment elevation
- B. ○ Isoelectric ST segments
- C. ○ T wave inversion
- D. ○ PR segment depression
- E. ○ Upsloping ST segment depression

Hide Explanation

The correct answer is A. ST segment elevation indicates acute coronary occlusion (and transmural infarction) in need of immediate reperfusion therapy. Isoelectric ST segments (B) are normal. While T wave inversion (C) it is relatively nonspecific and not a principal finding in acute coronary occlusion. PR segment depression (D) may be seen in pericarditis but is not a sign of acute coronary occlusion. While any ST segment depression can indicate ischemia, ST depression, particularly if it is upsloping, is nonspecific.

4. ST elevation in V1-V6, I, and aVL with ST depression in II, III, and aVF suggests occlusion of which artery?

- A. ○ Mid left anterior descending
- B. ● Proximal left anterior descending
- C. ○ Left circumflex artery
- D. ○ Right coronary artery
- E. ○ Diagonal branch

Hide Explanation

Proximal LAD occlusion (B) causes extensive ST elevation in V1-V6, I, and aVL with reciprocal ST depression in inferior leads II, III, aVF, reflecting large anterior STEMI. (A) Mid-LAD involves fewer leads without I and aVL. (C) LCX affects lateral leads. (D) RCA causes inferior changes. (E) Diagonal branch occlusion yields limited lateral involvement.

How would you rate this Brick?

References

- https://ecg.utah.edu/lesson/9 ECG learning center, section on myocardial infarction.
- Goldberger's electr lable at Clinical Key in the AWSOM library

References

- https://ecg.utah.edu/lesson/9 ECG learning center, section on myocardial infarction.
- Goldberger's electrocardiography, chapters 9 and 10, available at Clinical Key in the AWSOM library

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

- 01 ☑ ~~Describe and define the ST segment~~
- 02 ☑ ~~Recognize the localization and types of ST changes~~
- 03 ☑ ~~Compare common ST segment patterns and their implications for ischemia localization~~

Objective Confidence: 3/3